Chapter 21 – Numerical Differentiation

What's differentiation?

Differentiate means: mark by differences, distinguish

derivative: represents the rate of change of dependent variable with respect to an independent variable.

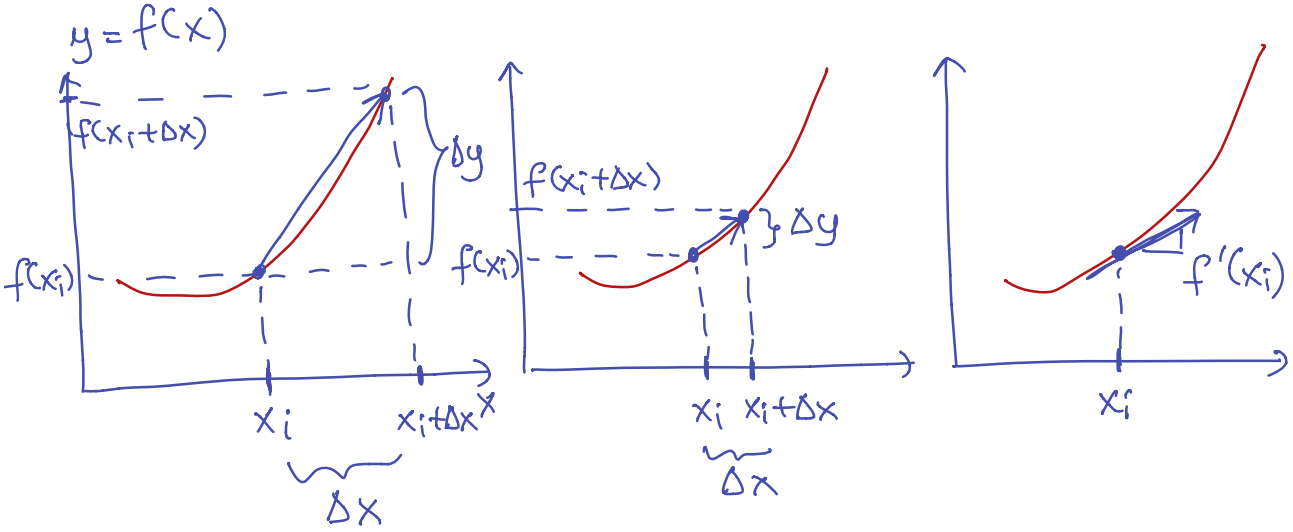
Graphical definition of derivative: Δx approaches zero, difference approximation becomes a derivative.

Difference Approximation: $$\frac{\Delta y}{\Delta x} = \frac{f(x_i+\Delta x) - f(x_i)}{\Delta x}$$

Derivative: $$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{f(x_i+\Delta x) - f(x_i)}{\Delta x}$$

- 1st derivative evaluated at x_i.
- slope of the tangent line at x_i.

$\frac{dy}{dx} = y' = f'(x_i)$ OR $\dot{y}$: used for time derivatives.



Second Derivative :

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\underbrace{\left(\frac{dy}{dx}\right)}_{\text{slope}}$$

tells us how fast slope is changing.

- curvature: high 2nd deriv. means high curvature

Partial Derivatives: $f(x,y)$

$$\frac{\partial f}{\partial x} = \lim_{\Delta x \to 0} \frac{f(x+\Delta x, y) - f(x,y)}{\Delta x}$$

partial deriv. of f w.r.t x at an arbitrary pt (x,y)

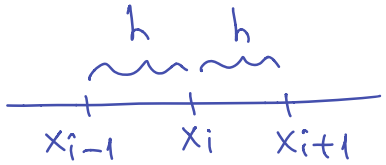
$$\frac{\partial f}{\partial y} = \lim_{\Delta y \to 0} \frac{f(x, y+\Delta y) - f(x,y)}{\Delta y}$$

High Accuracy Differentiation Formulas

Forward difference : $f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i} + O(h)$

$(h = x_{i+1} - x_i)$

Backward difference : $f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}} + O(h)$



Centered Difference: $f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1})}{2h} + O(h^2)$

forward Taylor series expansion:

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)}{2!}h^2 + \cdots \quad (1)$$

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h} - \frac{f''(x_i)h}{2!} + O(h^2) \quad (2)$$

Exclude second and higher order terms:

$$\boxed{f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h} + O(h)} \quad (3)$$

Forward difference approx.

Forward-difference approx. of 2nd derivative:

$$\boxed{f''(x_i) = \frac{f(x_{i+2}) - 2f(x_{i+1}) + f(x_i)}{h^2} + O(h)} \quad (4)$$

Insert (4) into (2):
collect the terms:

improved accuracy

$$\boxed{f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h} - \frac{f(x_{i+2}) - 2f(x_{i+1}) + f(x_i)}{2h} + O(h^2)}$$

Richardson Extrapolation

Use two derivative estimates to compute a third more accurate approximation.

(with step sizes h_1, h_2)

if $\boxed{h_2 = \frac{h_1}{2}}$

$$\boxed{D = \frac{4}{3}D(h_2) - \frac{1}{3}D(h_1)}$$

* For centered difference approx. with $O(h^2)$, this formula will give a new derivative estimate $O(h^4)$.

Ex Let $f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$

Evaluate the derivative at $x = 0.5$ using $h = 0.25$. (True value: $f'(0.5) = -0.9125$)

a) Forward, backward and centered diff.

b) Richardson extrapolation, with $h_1 = 0.5$, $h_2 = 0.25$

Soln

	Estimate	ε_t
Forward	-1.155	-26.5%
Backward	-0.714	21.7%
centered	-0.934	-2.4% ←

(b) $D = \frac{4}{3}D(0.25) - \frac{1}{3}D(0.5)$

$$D(0.5) = \frac{f(1)-f(0)}{1} = \frac{0.2-1.2}{1} = -1.0 \Rightarrow \varepsilon_t = -9.6\%$$

(centered diff; h_1)

$$D(0.25) = \frac{f(0.75)-f(0.25)}{0.5} = -0.9343, \quad \varepsilon_t = -2.4\%$$

(centered diff.)

Improved Estimate:

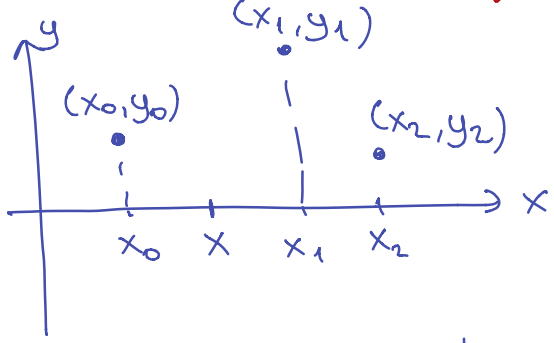
$$D = \frac{4}{3}D(0.25) - \frac{1}{3}D(0.5) = \frac{4}{3}(-0.9343) - \frac{1}{3}(-1)$$

$$\boxed{D = -0.9125}$$

this is exact, since $f(x)$ is a fourth order polynomial.

Derivatives of Unequally Spaced Data

Data from experiments



→ Fit a Lagrange interpolating polynomial to a set of adjacent pts that bracket the location (x) at which you want to evaluate the derivative

→ Differentiate the polynomial analytically and evaluate at x.

EX Fit a 2nd order Lagrange polynomial to 3 ptr.

$(x_0, y_0), (x_1, y_1), (x_2, y_2)$

$f(x_0)$ ↳ $f(x_1)$ ↳ $f(x_2)$

$$f(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} f(x_0) + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} f(x_1) + \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} f(x_2)$$

$$f'(x) = \frac{2x-x_1-x_2}{(x_0-x_1)(x_0-x_2)} f(x_0) + \frac{2x-x_0-x_2}{(x_1-x_0)(x_1-x_2)} f(x_1) + \frac{2x-x_0-x_1}{(x_2-x_0)(x_2-x_1)} f(x_2)$$

Derivatives and Integrals for Data with Error

* Data may have measurement error.
* Shortcoming of numerical differentiation: amplify errors in data

$$\frac{\Delta y}{\Delta x} = \frac{y_i - y_{i-1}}{\Delta x}$$ (Derivative of clean data)

$$\tilde{y}_i = y_i + \epsilon$$ ↳ small noise

$$\tilde{y}_{i-1} = y_{i-1} - \epsilon$$

Error amplification

$$\frac{\Delta y}{\Delta x} = \frac{(y_i+\epsilon) - (y_{i-1}-\epsilon)}{\Delta x} = \frac{y_i - y_{i-1} + 2\epsilon}{\Delta x}$$

* Error amplification occurs because differentiation is subtractive (Random positive and negative

errors tend to add)

* Integration is a summation process and + and − errors may cancel out.